The Real Estate Value Of Supermarket Endcaps

Why Location In-Store Matters

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Manufacturers pay premiums for endcap real estate because shoppers navigate the perimeter of stores, avoiding the aisles. Research has not established how the physical and visual reach of endcaps—product displays strategically placed at the end of a shopping aisle—might vary across locations in a store. This study explores how foot traffic and visual reach of endcaps differ by location. The most prominent endcaps, in terms of both foot traffic and visual reach, were at the back of the store. These had 24 percent more foot traffic and 30 percent more visual reach than front endcaps. Evidence from this study will help marketers reach more shoppers in different endcap locations in the supermarket.

INTRODUCTION

In-store media include any marketing effort or material, other than products, that communicates with shoppers in the supermarket (Sorensen, 2008). This category now accounts for a growing proportion of the marketing budget by brands and supermarkets in the United States, reaching 14 percent in 2015, up from 7 percent in 2012 (Angrisani, 2015). In Australia in 2014, in-store media represented a similar proportion of spending, at 16 percent, growing from just 6 percent in 2010 (Tonkin, 2014).

In-store media tend to be positioned in certain supermarket feature areas (Levy and Weitz, 2009), including entrances, endcaps, promotional aisles, and checkout areas, that are characterized by high foot traffic (*i.e.*, areas where a large number of shoppers walk) and visibility. Manufacturers aim to get their products into these areas (Buttle, 1993) to increase the probability of capturing customers' attention (Dulsrud and Jacobsen, 2009). Endcaps are the most common of these feature areas within the supermarket (Buttle, 1993; Levy and Weitz, 2009; Sorensen, 2008).

Endcaps are any displays for products located at the end of an aisle or row of shelving in a supermarket. They have become a common component of merchandising negotiations, because it is believed they guarantee higher shopper attention

Management Slant

- Challenging the industry myth that endcaps at the front of the store are the best, the results show that endcaps at the back of the store have more foot traffic and eyeballs than those at the front.
- Brands that use endcaps as an in-store promotion strategy should give preference to endcaps at the back of the store to maximize visual reach.
- Endcaps have a different amount of foot traffic and visual reach depending on location; thus, placement should be changed to reflect these differences.

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than shelves inside the main aisles (Point of Purchase Advertising International, 2012; Sorensen and Suher, 2010). Manufacturers value endcaps for the increased physical and visual availability they deliver for brands.

Retailers also value endcaps, historically using them to sell more merchandise, balance inventories, add excitement to the store, create low-price impressions, help reduce stock-outs on weekly feature advertisements, and offer shoppers suggestions on what to buy (Kline, 1960). The aim of all media is to reach as many potential customers as possible as close as possible to their potential purchase (Kennedy, McDonald, and Sharp, 2008). Placing products in multiple locations in the supermarket increases the chance that shoppers will notice, engage with, and purchase those products on the shopping trip.

The planning, selection, and setup of endcaps in different locations is an important activity for retailers and merchandising managers (Bogomolova, Szabo, and Kennedy, 2016). An estimated 30 percent of total supermarket sales come from endcaps (Sorensen, 2003). The competition for endcap space is so intense that retailers charge fees to manufacturers for placement there. Because of the proprietary nature of such negotiations, no one knows precisely how much retailers collect in such fees, but estimates range from $6 billion to $18 billion a year in the United States alone (O'Dwyer, 2015).

For marketers, understanding how to increase the reach of their brand to potential customers in store is essential for developing effective brand communication (Romaniuk and Sharp, 2015). Reach therefore is one of the most important components of a marketing strategy (Sharp, 2010). Shoppers cannot make a purchase if they cannot see or access an item. Few studies, however, have attempted to understand the location factors that could influence the reach of endcaps in a supermarket. The current research addresses that gap.

Whole-of-store research has found that shoppers follow predictable paths when navigating the supermarket (Fujino, Kitazawa, Yamada, Takahashi, *et al.*, 2014; Jung and Kwon, 2011; Larson, Bradlow, and Fader, 2005). These patterns affect what falls into the shoppers' field of vision (Sorensen, 2004). Industry experts speculate that one can gauge the value of endcaps on the basis of their proximity to the store entrance, with those closer to the entrance believed to have greater foot traffic and attention potential (Dapin, 2007).

In contrast, empirical studies of shopper navigation in store suggest that the back of the store has more foot traffic (Jung and Kwon, 2011; Larson *et al.*, 2005; Sorensen, 2004); it seems intuitive that endcaps here should have more visual reach, if more shoppers are in this area of the store. Such contrasting ideas can confuse brand managers and retailers making merchandising decisions, such as which endcap location to secure in negotiations. This research aims to empirically clarify the issue and resolve the confusion.

Numerous studies have asserted that the location of an endcap or display affects the sales uplift of products placed on them (Chevalier, 1975; Inman, Winer, and Ferrano, 2009; Nakamura, Pechey, Suhrcke, Jebb, *et al.*, 2014). How endcap sales affect size estimation is all the more challenging, however, because not all studies have attempted to understand the effect of display location and the resulting foot traffic and visual reach in the store. One study mentioned that a standard location was used for the two endcaps in each store (Chevalier, 1975); another made no mention of where these endcaps were located in the store (Kline, 1960). Another group of researchers controlled for the total number of promotions at different locations in their study but not the location of the endcap in the store (Nakamura *et al.*, 2014).

The first work to give an understanding of exposure to endcap displays in different locations in the store was in 2008 (Sorensen, 2008). The author tracked more than 20,000 shoppers and showed that there was a difference between the location of endcaps that got very little exposure and those that received a lot of exposure. The author defined "exposure" as when an endcap falls into the shopper's field of vision. That research, however, was conducted only in one country and used a split-aisle store format, which is different from the dominant supermarket format of long aisles found in most western countries.

The purpose of the current article is to measure whether the endcap pattern identified in that prior study (Sorensen, 2008) holds across the United States, Australia, and New Zealand and across a more prevalent store format. This research aims to extend the understanding of foot traffic and visual reach in the context of endcaps in the supermarket in three ways:

- It establishes how foot traffic past endcaps at different locations in the supermarket might differ. This provides an understanding of how customers shop the supermarket.
- It identifies which endcap locations appear in the shoppers' field of vision (visual reach). This provides marketers with insights into which endcap locations to target for their brands to achieve the most visual reach.
- The research explores when shoppers walk past and when they actually look

at endcaps over the course of their shopping trip.

Findings from these three research objectives will allow retailers and manufacturers to understand which locations in the store could help generate the most return on merchandising investment.

LITERATURE REVIEW

Supermarket Shopper Paths

Many retailers and marketers traditionally have assumed shoppers methodically walk up and down each aisle and then go to a checkout to end their trip. Empirical observations, however, show that shoppers do not go up and down every aisle in the supermarket; instead, they shop the perimeter of the store and make short excursions into the aisles to get what they need, then return to the perimeter of the store and end their trip as soon as they obtain all the items they are looking for (Jung and Kwon, 2011; Larson *et al.*, 2005; Sorensen, 2004). As a result, most shoppers cover less than a quarter of the floor space of a store on a shopping trip (Bogomolova, Anderson, Kennedy, Page, *et al.*, 2013), and fewer than two percent cover more than half the store (Scamell-Katz, 2012).

Given what is known about the high degree of preplanning for supermarket shopping, this task-oriented behavior in navigation might be expected. Shoppers do not enter the aisles or go into an area if they do not need to buy something there. The majority of shoppers are very familiar with their store (Scamell-Katz, 2012), and because many only want to purchase a few items (Sorensen, 2016), most do not shop the entire store.

The longer a shopper spends in a store, however, the greater is the chance that he or she will enter the aisles (Sorensen, 2009). This also suggests that items placed in the middle of an aisle do not get as much attention as those on the ends, because people complete their task and return to the perimeter before reaching the midpoint. Because of these patterns, the shopper literature often has stated that shoppers more likely will purchase products displayed at the end of an aisle (on endcaps) than in aisles because of greater traffic around the perimeter of the store (Inman *et al.*, 2009).

Additionally, given that many shoppers make small trips to a supermarket to get just a few items on any single visit (Sorensen, 2016), the likelihood of all shoppers reaching all the aisles, even at the endcap, is reduced; many end their trip earlier and return to the front of the store for checkouts. This suggests that endcaps located in different areas within the store—especially in relation to being closer or further away from the entrance—should have a different number of shoppers walking past them. The first research question therefore was as follows:

RQ1 How does foot traffic past an endcap differ by the location of the endcap, in particular those located at the front versus the back of the supermarket?

Reach in the Supermarket

Supermarkets serve not only as a retail outlet but as an advertising channel for manufacturers (Sorensen, 2008; Woodside and Waddle, 1975). A typical supermarket stocks 30,000 to 50,000 different stock-keeping units, and a large supermarket stocks up to 150,000 different stock-keeping units (Krafft and Mantrala, 2010). Brand managers always are looking for ways to increase the exposure of their brands to be considered for purchase against a multitude of competitors (Egol and Vollmer, 2008).

A shopper's attention consequently is one of the most important resources to consider for in-store marketing decisions (Corstjens and Corstjens, 1995; Wedel and Pieters, 2008). Academic studies describing the purchase potential of in-store media-use attention (visual reach) as a baseline metric to help understand the sales effectiveness of different media (Nordfält, 2011; Sorensen, 2008). The understanding is that the more shoppers a display reaches, the more the sales increase. In line with this thinking, brand managers aim to place their products in these high-attention locations.

Capturing attention is necessary but not sufficient for customer purchase (Nordfält, 2011). Numerous studies have shown that other factors, including price (Chandon, Hutchinson, Bradlow, and Young, 2009; Sorensen, 2010), quality (Sivakumar, 1995), pack size (Yan, Sengupta, and Wyer, 2014), nutritional information (Cowburn and Stockley, 2005; Enright, Good, and Williams, 2010), and product benefits (Lai, 1995; Meyvis and Janiszewski, 2002) affect purchase decisions. A product cannot be purchased if it does not fall into shoppers' field of vision, however (Sorensen, 2003; Varela, Antúnez, Cadena, Giménez, *et al.*, 2014; Wedel and Pieters, 2007).

A previous study attempted to understand visual exposure of endcap displays in store, using a field-of-vision tracking method (Sorensen, 2008). The author found that exposure to endcaps followed the paths shoppers took in the supermarket, and endcaps more often were in the shoppers' field of vision than were other types of in-store media (Sorensen, 2008). A major limitation of the study was its scope of just one store, a smaller supermarket that used a split-aisle format with generally shorter than average shopping trips—12 minutes versus 22 minutes, a benchmark developed through a meta-analysis across many supermarkets (Bogomolova *et al.*, 2013). It is therefore important to test whether the pattern observed can be replicated with multiple

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observations across supermarkets and countries to understand the generalizability of the findings and boundary conditions (Ehrenberg and Bound, 1993; Macdonald and Sharp, 2000).

On the basis of what is known about shopper navigation in the supermarket, endcaps in different locations should have a different number of shoppers walking past them. As a result, the visual reach of endcaps across the shopper base also should be different. This led to the second research question:

RQ2 How does visual reach, in terms of endcaps in the shopper's field of vision, differ by the location of the endcap, in particular those located at the front versus the back of the supermarket?

Reaching Shoppers First

Supermarket shopping is considered a low-involvement activity that occurs in a familiar environment (East, Hammond, Harris, and Lomax, 2000; Nenycz-Thiel and Stern, 2012; Uncles and Kwok, 2008). Decisions are made within seconds (Dickson and Sawyer, 1990; Hoyer, 1984; Shiv and Fedorikhin, 1999), and the chosen product is often the first one customers see (Carney and Banaji, 2012). It therefore is not surprising that brands try to minimize interference from competitors' products and communications and to make their products stand out as much as possible in this familiar environment (Woodside and Uncles, 2005).

In-store activities play a significant role in this effort, given their potential to divert shoppers from their planned brand purchases (Abratt and Goodey, 1990; Huang, Hui, Inman, and Suher, 2012). Previous research has shown that between 37 and 67 percent of shoppers use a shopping list (Inman *et al.*, 2009; Manno, Dunn, and Bogomolova, 2015; Thomas and Garland, 2004), which suggests that a high percentage of purchases in store are planned. This planning, however, happens more at a category level (87 percent) than at a brand level (7 percent). Very few shoppers actually plan to buy specific brands, and, of those who do, approximately one-third actually change brands (Manno *et al.*, 2015).

Increasing the visual reach of brands compared with the competition thus could nudge shoppers to buy those brands in the supermarket if they are looking to purchase from the category (Kalnikaitė, Bird, and Rogers, 2013). The most common in-store feature area is endcaps, which can act as a billboard, a reminder, and a prompt to purchase (Finne and Sivonen, 2008; Underhill, 1999). Marketers know endcaps expose shoppers to products in a space free from direct aisle-based competition (Dulsrud and Jacobsen, 2009; Schindler, Berbaum, and Weinzimer, 1987), and reaching shoppers first in this way thereby could increase the likelihood of a purchase occurring (Carney and Banaji, 2012).

The authors considered when shoppers walk past endcaps at different locations and when those endcaps appear in the shopper's field of vision during the course of his or her shopping trip. No published study to date has looked at when in the shopping trip shoppers walk past endcaps and which endcap (front versus back location) might appear in the shopper's field of vision in the supermarket. The third research question thus was posed:

RQ3 Over the course of a shopping trip, when (earlier or later) do shoppers walk past endcaps, and which endcaps (front versus back) do they walk past and look at?

METHODOLOGY

Data and Sampling

To address the research questions, the authors used the most rigorous empirical data source: direct observations in real-life supermarkets, using field-of-vision recorders. The sample consisted of 169 real-life, unscripted shopping trips across three countries (United States, Australia, and New Zealand) in five suburban stores:

- Store 1: Australia, $n = 107$;
- Store 2: Australia, $n = 25$;
- Store 3: United States, $n = 5$;
- Store 4: United States, $n = 11$;
- Store 5: New Zealand, $n = 21$.

The researchers obtained agreements with store management to gain permission for observations. The primary data collection for this study (Store 1) was conducted with Tobii Eye Tracking Glasses. This was complemented by secondary field-of-vision data (Stores 2–5) from past commercial shopper behavior studies by TNS, deidentified and adapted to academic purposes. The demographic breakdown of Store 1 ($n = 107$) was 70 percent female, which is consistent with that of previous studies that have shown that women still do approximately two-thirds of the grocery shopping (Bogomolova, Vorobyev, Page, and Bogomolov, 2016).

Participants were recruited by convenience-sampling techniques; shoppers were intercepted *in situ* as they entered each supermarket. The sample was composed of shoppers who were going to do their usual shopping and were not coming to the store for the purpose of participating in market research. No instructions about where to go or what to buy were given to participants. The participants paid their own money for all purchases made on the trip.

The video footage was recorded between December 2008 and December 2015. A total of 56 hours of video were analyzed. The mean shopping time was 20 minutes (*Mdn* = 17 minutes), which is similar to the mean shopping time of 22 minutes (*Mdn* = 18 minutes) found across the same three countries in a previous shopper study of 86,700 shopping trips across 17 supermarkets (Bogomolova *et al.*, 2013).

Procedure and Observational Measures

All five stores analyzed had a similar layout, with no middle aisle and an entrance and checkouts at the front. This is representative of the format found in the majority of supermarkets in these countries. Endcaps were labeled as either front or back of store, depending on their location.

Data were collected through specially designed glasses that recorded shoppers' field of vision; these included Tobii Eye Tracking Glasses and glasses provided by TNS. Both glasses consisted of a built-in camera and microphone and a small bag containing a recorder, storage, and battery pack. In all studies, shoppers were aware that what they were looking at in store was being recorded, although they were not told that endcaps were of interest in the study.

The measures used in the analysis were an adaptation of the metrics that were developed in line with the path-to-purchase model (Sorensen, 2003). These metrics have been used in the past to determine how influential in-store media are at affecting shopper behavior (Nordfält, 2011; Sorensen, 2003).

The metrics below were used to determine the foot traffic (RQ1) and visual reach of endcaps (RQ2). Total traffic (number of shoppers) in the store was used as a baseline (Nordfält, 2011). This provides a more comparable measure than only counting the ratio of the shoppers passing a display (as in Sorensen's, 2003, study). It allows for an overall understanding of how each endcap location compares in regard to the total store traffic.

The traffic ratio is expressed as a measure of the number of shoppers who walk past an endcap within an arm's distance (approximately 1 meter) as a ratio of the total number of shoppers who enter the store. The traffic ratio helps to determine where high-traffic locations are in the store.

$$\frac{\text{Total number of shoppers who walk past an endcap within 1 meter}}{\text{Total number of shoppers who enter the store}} \quad (1)$$

The visual-reach ratio is expressed as a measure of the number of shoppers who walk past an endcap within an arm's distance and look (fixate their eyes) on an endcap, as a ratio of the total number of shoppers who enter the store. The reach ratio helps determine where high-visual-attention locations are in the store. A "look" has been defined in this research as more than a glance; it requires a shopper to make eye contact with a display for an adequate period of time, just under a second (Nordfält, 2011).

$$\frac{\text{Total number of shoppers who walk past an endcap within 1 meter and look at the endcap}}{\text{Total number of shoppers who enter the store}} \quad (2)$$

Recordings were coded by interaction with the endcaps (traffic or visual reach), which represented the units of interest. Variables describing the length of trip were recorded separately. The researchers recorded the starting time of each interaction to make returning to the interaction for verification possible. This also allowed for an understanding of at what point in the shopping trip (start, middle, or end) shoppers were walking past and looking at endcap displays (RQ3). A total of 5,548 interactions were recorded.

Three coders watched examples of the interactions with the lead researcher on the project. Collaborating closely, the coders then coded their assigned videos, consulting the lead researcher when they encountered questions of interpretation. The intercoder reliability of three shopping-trip videos coded by three researchers for all measures was .89 (Cohen's kappa), which is an acceptable level of agreement for this type of coding (Lombard, Snyder-Duch, and Bracken, 2004).

The authors performed independent-sample t tests on gender for the average time spent in store, endcaps walked past (traffic ratio), and endcaps in the shopper's field of vision (reach ratio) to understand how patterns might differ between shoppers of different genders and ages. The primary analysis involved cross-tabulations between the endcap location (front versus back) and ratio measures (traffic and reach), with Pearson's chi-square statistics employed to determine the statistical significance of the result. The patterns in the ratios examined were of interest rather than statistical significance alone. Where appropriate, weighted averages based on the sample of shoppers were used to describe the overall pattern, because not all stores had the same number of shopper videos.

RESULTS

The results are presented and discussed by research question. First, the authors

explain how foot traffic past an endcap differed by the location of the endcap in the supermarket (RQ1). Next they document how visual reach (the endcap in the shopper's field of vision) differed by the location of the endcap in the supermarket (RQ2). Finally, the authors examine at what point shoppers walked past and looked at endcaps over the course of their trip to better understand when endcaps at different locations are viewed (RQ3). Where appropriate, they discuss whether any demographic differences (gender or age) were present. This analysis was done only with the primary data-collection sample of shoppers (n = 107).

TABLE 1

Endcap Traffic (RQ1) and Reach (Q2) Ratios (%)

Traffic Ratio	Store 1	Store 2	Store 3	Store 4	Store 5	Average
Front (%)	68	55	57	52	62	64
Back (%)	86	89	92	85	99	88
Diff front vs. back (%)	−19*	−34*	−34	−34	−37*	−24*
Visual Reach Ratio	**Store 1**	**Store 2**	**Store 3**	**Store 4**	**Store 5**	**Average**
Front (%)	25	23	22	17	20	24
Back (%)	52	55	57	55	55	54
Diff front vs. back (%)	−27*	−32*	−35	−38	−35*	−30*

Note: Diff = difference between.$p < .05$.*

TABLE 2

Endcap Traffic and Visual-Reach Ratio Differences

Weighted Average of Five Stores	Traffic Ratio (%)	Visual Reach Ratio (%)	Difference (%)
Front	64	24	−40
Back	88	54	−34

RQ1: Foot Traffic Past Endcaps At Front versus Back Locations

The researchers first examined foot traffic differences by location of the endcap in the supermarket for the five stores (See Table 1). Overall, using weighted averages, there was 24 percent more foot traffic past endcaps at the back of the store compared with the front (64 percent front, 88 percent back). The interstore variation for this difference ranged from a 19 percent difference in Store 1 to a 37 percent difference in Store 5. Overall, endcaps at the back of the store had, on average, about one-quarter more foot traffic than those at the front.

The authors found no significant difference in endcap traffic between the male (M = 16.69, SD = 6.43) and female (M = 18.52, SD = 8.28) shoppers, $t(106) = -1.12$, p = .265. This indicates that all shoppers used the outside of the store as the main pathway, which is not surprising. The authors found that younger shoppers (ages 18 to 49 years; M = 20.14, SD = 7.53) walked past more endcaps (four more, on average) than older shoppers (M = 16.19, SD = 7.61), $t(106) = 0.695$, p = .008.

This difference could suggest two things. Younger shoppers might have visited more of the store and so walked past more endcaps. Alternatively, older shoppers might be more effective at shopping and know where items are placed; as a result, they would not need to walk greater distances in the store.

RQ2: Visual Reach of Endcaps For Front and Back Locations

Next, the authors explored visual reach ratios by location of the endcap in the supermarket (See Table 1). Overall, there was 30 percent more visual reach at the back of the store compared with the front (54 percent at the back, 24 percent at the front). This ranged from a 27 percent difference in Store 1 to a 38 percent difference in Store 4. The answer to RQ2 is that endcaps at the back of the store had, on average, about one-third more visual reach compared with those at the front.

In three of the five stores observed, these differences for both traffic and visual reach ratios were statistically significant at $p < .05$. The other two stores trended in the same direction, showing a similar pattern between front and back endcaps for RQ1 and RQ2. Overall, the differences for traffic and visual reach were statistically significant at $p < .05$.

To understand the cumulative difference between traffic and visual-reach ratios of endcaps at the front and back of the store, the authors compared them (See Table 2). This difference was 6 percent (40 percent front, 34 percent back). This indicates that endcaps located at the back of the store visually reached 6 percent more shoppers who walked past them than those at the front of the store. To build reach on endcaps, brand managers have a slight advantage if they use endcaps at the back of the store. Not only does more foot traffic pass there, but these endcaps are seen by slightly more shoppers walking past.

The authors found that there was a significant difference in the endcap reach score for the male (M = 7.63, SD = 4.72) and female (M = 10.15, SD = 5.37) shoppers, $t(106) = -2.31$, $p = .023$. This result should not be surprising, given that women are more efficient shoppers than men (Bogomolova, Szabo, and Kennedy, 2016). To be more efficient at shopping, women likely pay more attention to products on shelves; this includes endcaps.

The authors found no significant difference in endcap visual reach between younger (M = 10.01, SD = 5.03) and older shoppers (M = 8.88, SD = 5.47), $t(106) = 0.885$, $p = .267$. This suggests that younger shoppers paid less attention to products on endcaps, given that they walked past more endcaps (four more, on average) than older shoppers.

RQ3: Shoppers Looking at Endcaps During Their Shopping Trip

The authors broke shopping trips down into deciles in terms of time, to give an indication of when during the shopping trip shoppers walked past and looked at endcaps in different locations (front versus back). Overall, traffic and visual-reach ratio occurrences over the first 90 percent of the trip duration showed a similar pattern; when shoppers walked past endcaps, they likely looked at them. In the last 10 percent of a shopping trip, visual reach dropped compared with traffic, which suggests that shoppers were walking past endcaps still but were not looking at what was placed on them. They likely were occupied by the task of getting to checkouts and out of the store, having completed their shopping tasks.

To further understand the pattern identified above, the authors graphed the location of endcaps that shoppers walked past and looked at over the course of their trip (See Figures 1 and 2). Among participants who were in the store for 10 minutes or less, 28 percent of their traffic came from endcaps at the back of the store (See Figure 1).

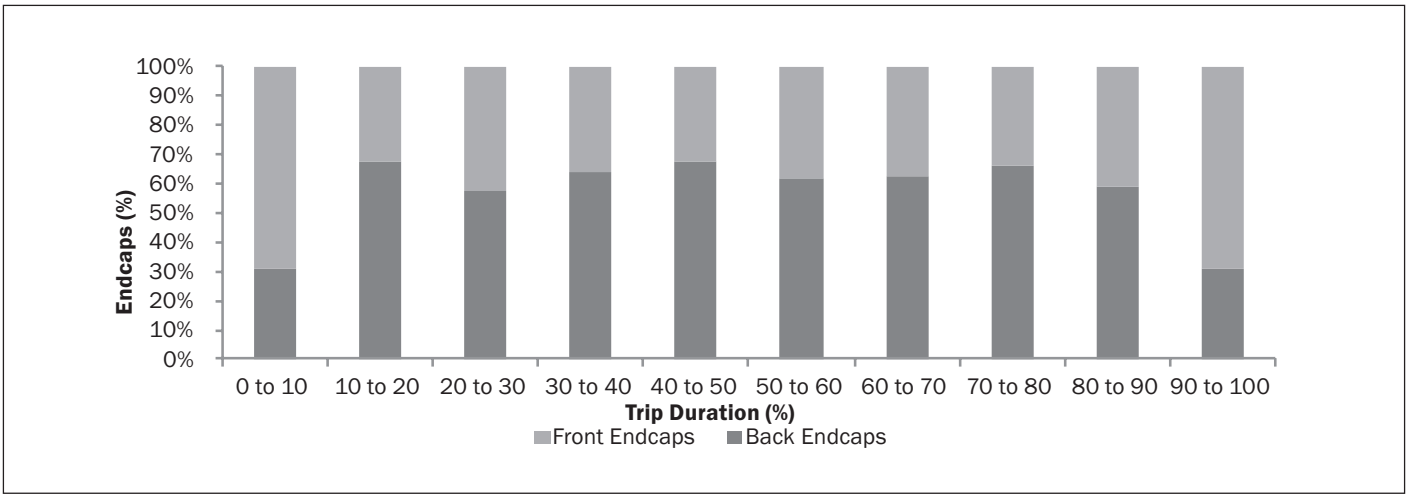


Figure 1 Endcap Traffic Ratio Front versus Back Comparison during a Shopping Trip

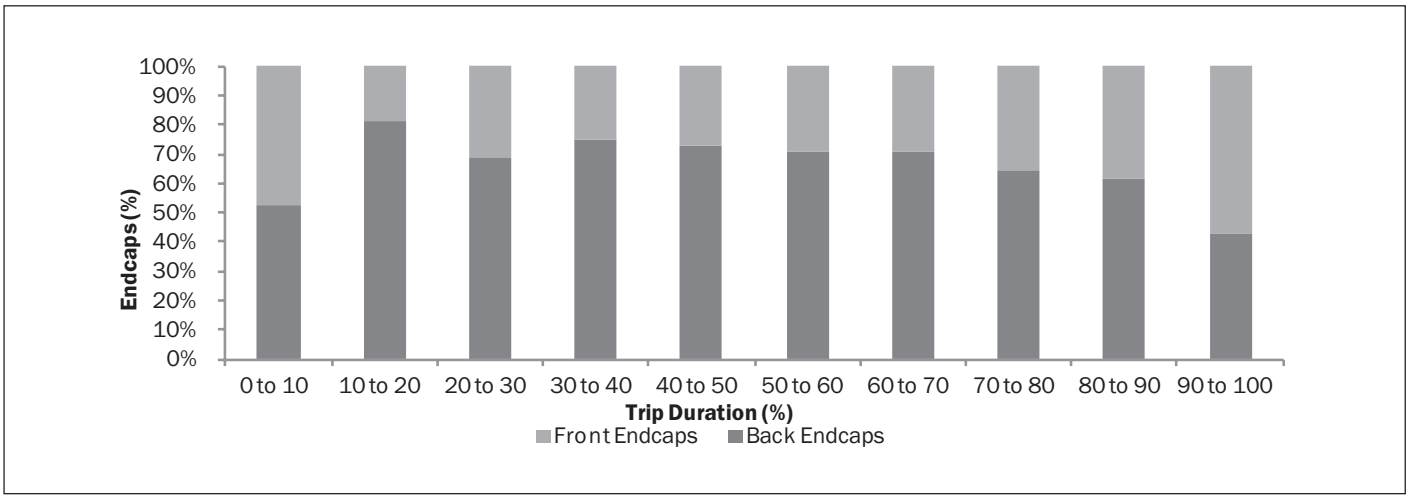


Figure 2 Endcap Visual-Reach Ratio Front versus Back Comparison a Shopping Trip

Overall, shoppers walked past endcaps at the back of the store more, and these endcaps appeared in the shopper's field of vision more than endcaps at the front, except in the first and last 10 percent of a shopping trip. This provides evidence that people use the back of the store more during shopping trips for navigation between aisles. Traffic at the front of the store is only greater in the first and last 10 percent of shopping trips, because shoppers either are entering the store or are exiting at these times.

Time Spent in Store

The authors found that men (M = 15.65, SD = 11.54) spent significantly less time (eight minutes) in the store than women (M = 23.52, SD = 12.52), $t(106) = -3.05$, $p = .003$. This is consistent with past research that suggests that women spend more time grocery shopping than men (Blaylock and Smallwood, 1987; Bogomolova, Vorobyev, Page, and Bogomolova, 2016; Sommer, Wynes, and Brinkley, 1992). The authors found no significant difference in time spent in the store between younger (M = 22.72, SD = 12.91) and older shoppers (M = 19.92, SD = 12.51), $t(106) = 0.458$, $p = .256$.

This suggests that younger shoppers had a quicker walking speed compared with older shoppers, given that they walked past more endcaps but spent a similar amount of time in the store. This is consistent with previous work (Sorensen, 2003). It appears that the quicker walking speed of younger shoppers comes at the expense of paying attention to products that are around them, in this case on endcaps, because there was no difference in endcap reach for younger and older shoppers.

DISCUSSION

Using observations of real, *in situ* shopper behavior across three countries and five supermarkets, this study develops new knowledge regarding the effectiveness of endcaps as in-store media. The authors identified which locations of endcaps in the store (front versus back) attracted higher shopper traffic and reach. This knowledge enhances the body of evidence regarding the efficiency of endcaps, allowing manufacturers and retailers to make better informed decisions about in-store marketing and shelf allocation.

This study found that endcaps located at the back of the store had, on average, 24 percent more foot traffic and 30 percent more visual reach compared with endcaps located at the front.

This study found that endcaps located at the back of the store had, on average, 24 percent more foot traffic and 30 percent more visual reach compared with endcaps located at the front. The traffic results are consistent with earlier path-tracking studies (Larson *et al.*, 2005; Sorensen, 2009). Both of these prior studies indicated that the back of the store should have more foot traffic than the front, on the basis of the paths shoppers took in the store—yet they did not examine specifically by how much. The present study quantifies the finding that endcap locations have different amounts of traffic and visual reach in the supermarket. Endcap observations in this research provide further empirical support and benchmarks for these previous findings.

The research also adds to the literature highlighting the importance of understanding how shoppers behave in the supermarket and the navigation routes they take. By determining which endcaps have the most traffic and visual reach, the findings inform marketing theory and practice about possible return on investment into different ways to build physical availability in the supermarket.

Implications for Practitioners And Researchers

This article provides a number of contributions. First, it extends display research and provides a straightforward research method and associated set of metrics. Second, the results provide an indication of what to expect in terms of foot traffic and visual reach in the supermarket. These contributions have direct implications for retailers and manufacturers, because this better knowledge of endcap-display locations aids in decision making about this medium.

For researchers, the results give a descriptive understanding of the traffic difference past endcaps and the visual attention of each, on which future studies can be based. The current study allows practitioners to understand the importance of location in the supermarket and the accompanying visual reach when performing display-effectiveness studies. Researchers need to control for display location in the store by recording the number of shoppers who walk past and look at displays (Nordfält, 2011).

For retailers, the study gives an indication of which endcap space in the store likely will generate the most return on merchandising investment. Understanding how to reach customers in store is crucial for developing an effective brand communication (Romaniuk and Sharp, 2015).

This result is not necessarily intuitive; for example, retailers may be tempted to think endcaps at the front of the store have the best potential for merchandising return on investment (high reach) given their proximity to the entrance (Dapin, 2007). The results here suggest that brand strategies that aim to place products on endcaps closer to the entrance might not be justified, because their purported superior effectiveness is not supported by evidence. Although the choice to place brands on displays at the front of the store still might have a positive return on investment compared with simple placement in an aisle, more shoppers walk past and look at endcaps at the back of the store compared with the front.

Overall, shoppers make decisions quickly in store. Visual reach in a crowded environment is critical. There is still a strong rationale for brand managers to place products on endcaps, because these areas are free from aisle-based competition, whether at the front or the back of the store. Endcap locations do vary in terms of the number of shoppers walking past them and looking at them, so understanding where to reach more shoppers is of benefit to both retailers and brand managers.

Limitations and Future Research

The main limitation of the research design relates to the use of secondary data, which were collected in commercial research studies for analysis in Stores 2–5 and did not have any accompanying demographic information about shoppers. This only allowed a demographic analysis on the shoppers (n = 106) in Store 1. Although using secondary data provided a larger dataset and enabled results to be generalized across varying conditions, the study is constrained to the data of the previous studies. The consistency of the results across stores in spite of this limitation highlights the robustness of the findings.

The second limitation is that the study only included the stores with the most common supermarket format (*e.g.*, entrance to one side, and aisles of products). Future researchers could explore other formats, including different locations of checkouts or configuration of aisles, to see whether the patterns hold.

Next, the study focused on supermarket grocery stores only. Researchers could analyze other stores, such as department or hardware stores, to see whether similar patterns of foot traffic and visual attention occur. Insights into the extent to which foot traffic and visual-reach metrics might differ between endcaps and products in the aisles would be of interest. This information would help researchers to understand shopping-path behaviors and the number of shoppers entering the aisles. Last, the study could be expanded to include more shoppers, so a clearer pattern of the order in which endcaps appear in the shoppers' field of vision could be obtained.

Another potentially useful area of future research is to combine the approach used here (with its focus on traffic and visual reach) with technological advances in purchase data or stock taking of items on endcaps. This information could be used to ascertain the extent to which shoppers' purchases from endcaps reflect the same patterns as observed through foot traffic and visual attention. This study found that shoppers looked more at items placed on endcaps at the back of the store, but understanding the relationship this pattern has with sales will have greater impact on the return-on-investment metrics and therefore is an important area of future research.

That said, understanding endcap-sales uplifts involves more than the variable of location. This research provides insights on how to reach the most shoppers, but researchers will need to control other variables, such as price (promotional depth), quality, pack size, category, and advertising support, in any sales analysis to determine the return on investment for endcap locations in the supermarket.

Research has found that product locations in the supermarket can influence search and browsing times in different areas of the store (Pinna and Seiler, 2014). The current authors believe the type of data and approach of this research open the door to future, more detailed studies of product location, product sales, and store design that will help brand managers get more return on investment from their in-store advertising expenditure. JAR

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